# 1. Standard Models vs Reasoning Models

## 1.1 Standard (Non-Reasoning) Models

Standard LLMs generate responses directly from the input prompt.

**Characteristics:**

- Thinking and answer generation occur simultaneously.
- No explicit planning before responding.
- Fast and cost-efficient.
- Suitable for simple queries, summarization, translation, and general conversation.

**Limitations:**

- Weak multi-step reasoning.
- More prone to logical errors and hallucinations.
- Early mistakes can propagate (snowball effect).

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## 1.2 Reasoning Models

Reasoning models explicitly separate the thinking phase from the final answer generation.

**Characteristics:**

- Perform internal analysis before producing the output.
- Explore multiple reasoning paths.
- Validate logic before responding.
- More reliable for complex tasks (math, coding, planning).

**Trade-offs:**

- Higher latency (slower responses).
- Higher computational cost.

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# 2. Plan-and-Solve Prompting

## 2.1 Definition (Original)

**Plan-and-Solve prompting is a structured prompting technique that forces a language model to first create an explicit step-by-step plan for solving a problem and only then execute that plan to produce the final answer.**

It artificially separates planning from execution, allowing even standard models to behave more like reasoning models.

Eg:-

Let's first understand the problem and devise a plan to solve the problem.
Then, let's carry out the plan and solve the problem step by step.

Do not directly solve the problem , first understand the problem , devise an plan to solve the problem , extract relevant variables and their corresponding numerals, while creating plan form intermediate steps and then carry out the plan step by step(while solving the intermediate steps , pay attention to calculations and common sense), solve the plan step by step.

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## 2.2 Why It Works

Standard models often struggle with complex problems because reasoning and answer generation occur simultaneously, leaving little opportunity for structured analysis or verification. **Plan-and-Solve prompting** addresses this limitation by introducing a deliberate planning stage before solution generation. This approach:

- Enforces an explicit problem-decomposition phase prior to answering
- Reduces logical and procedural errors by clarifying the solution path
- Improves coherence, structure, and overall accuracy of responses
- Mitigates hallucinations, especially in multi-step or complex tasks, by grounding the answer in a predefined plan
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## 2.3 Methodology

**Phase 1 — Planning**
The model is instructed to:

- Understand the problem
- Identify variables and constraints

- Determine formulas, rules, or steps required
- Produce a clear solution strategy

**Phase 2 — Solving**
The model:

- Executes the plan step-by-step
- Produces the final answer based on the plan

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# 3. Industry Use Cases of Plan-and-Solve Prompting

## 3.1 Customer Support Systems

Used to ensure structured, accurate responses:

1. Analyze user query and intent.
2. Retrieve relevant information.
3. Plan the response strategy.
4. Generate the final reply.

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## 3.2 Code Generation & Debugging Tools

Plan-and-Solve helps models:

- Understand requirements
- Break problems into modules
- Generate structured code
- Debug step-by-step

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## 3.3 Agentic AI Systems

Autonomous AI agents use planning before actions:

- Task decomposition
- Tool selection
- Execution sequence
- Final output synthesis

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## 4. Key Takeaways

- Standard models are fast but less reliable for complex reasoning.
- Reasoning models are accurate but expensive.
- Plan-and-Solve prompting enables standard models to perform structured reasoning.
- Widely used in production AI systems for reliability and control.

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## 5. Conclusion

Plan-and-Solve prompting is a practical and powerful technique that bridges the gap between standard and reasoning models. By enforcing an explicit planning stage before execution, it significantly improves accuracy, transparency, and robustness of LLM outputs in real-world applications.